

ENVIRONMENT AND ACCESS TO RESOURCES IN AFRICA

Piers Blaikie

This article has been written as a contribution to the future orientation of a research programme on the agrarian crisis in Africa, which has been set up by the Joint Committee on African Studies (JCAS) of the American Council of Learned Societies and the Social Science Research Council. The aim of this article is to provide an agenda for research on the environment and access to resources in Africa and is one of four which both provide a review of some of the most important research issues and suggest ways in which they might be tackled.

Four premises underlie this research initiative and these have been expressed in an earlier article (JCAS, 1986a: 3). These are: (a) that African agriculture can be understood in terms of a series of technical operations with a variety of biological, social, political and ecological processes acting upon each of them; (b) that farming structures can only be understood in relation to overall patterns of resource control; (c) that political, economic and cultural structures in Africa are continually changing as they become more integrated into national and international systems; and (d) that the functioning of households, regional economies and national and international systems are interdependent (*ibid.*: 3). These premises, which are referred to below, can be summarily labelled the interaction/interdisciplinary premise; the dynamic control of resources premise; the incorporation/integration premise; and the levels and scales premise. Each of them is explicitly examined below with particular reference to the environment and access to resources.

In the light of these premises the subject 'environment and access to resources' is interpreted as follows. There is a dynamic relationship between patterns of access to resources and the 'environmental data' on which the outcomes of patterns of access are played out. These outcomes take the form of specific land uses, the use and movement of soil nutrients and vegetation, of investments or dis-investments in the capability of land and so on. Therefore these outcomes modify the environmental data so that future patterns of access both adapt to and further change the environment in a recursive and dialectical manner. Of course these outcomes must not be seen only in environmental terms, but also as processes of the agrarian political economy itself. Livelihoods are secured or lost, some people accumulate capital and some lose it, differentiation of the rural population may proceed or be checked, and so on. None the less, it is essential to link agrarian and environmental change firmly together in theory and in empirical work.

The article starts with the proposition that many issues of environment and access to resources are clouded with uncertainty. This arises out of a lack of reliable data, particularly on environmental change; ideological and technical problems of definition and measurement; the inability to identify reliably human agency in environmental change (rather than, for example, climatic change); a wide range of contradictory interpretations of the impact of land degradation (Does it matter? To whom? Why?); different levels of

abstraction and scale involved in analysis; and the 'atomisation' of theories linking environmental and social change. These sources of uncertainty are briefly rehearsed, and then a set of responses in terms of future research agenda is proposed. A central part of this set is a dynamic resource access-environmental change model linked to a decision-making model which focuses on land users' responses to environmental change. These two models provide a clearly specified context and anchoring point for a number of theoretical issues in African agrarian studies which hitherto have not been linked together.

UNCERTAINTY IN RESEARCH ON ENVIRONMENT AND ACCESS TO RESOURCES

There is an enormous range of opinion about many aspects of social and environmental change. Perhaps the most uncertain aspect is the extent to which environmental deterioration, specifically land degradation, is frequently seen to be a crucial factor in the African agrarian crisis. In the purely 'social' (i.e. political, economic, sociological) sphere of agrarian change there is a great deal more written and less uncertainty, although as JCAS (1986b) has summarised, there remain ambiguities and an enduring impression that most hypotheses prove to be conditional ones only. Uncertainty in the political, economic and sociological spheres arises not so much as a result of data and measurement problems and/or ideological disagreement, but because of an extraordinary 'atomisation' of theories of agrarian change, and a failure to link them to the environment/social dialectic outlined above.

In the area of understanding and evaluating environmental degradation in Africa, the following causes of uncertainty emerge. First, there is the problem of data—its scantiness, unreliability, irrelevance and ambiguity. 'Statistics are seldom in the right form, are hard to come by, and even harder to believe let alone interpret' (United Nations, 1977: 177). There is probably a number of areas in Africa where, unfortunately, the certainty of the problem of land degradation is borne in upon local residents and outside commentators all too clearly. The case of Ethiopia is an obvious one, that of overwhelming evidence of erosion along with low levels of productivity, poverty, a long history of extractive semi-feudal social relations of production and the superimposition of exceptionally dry years, all combining to produce chronic famine (Uhlig, 1984). Even here, though, the degree to which erosion is responsible in terms of spatial variability and its contribution to low and variable yields of basic foodcrops and livestock can only be guessed at. The recent Food and Agricultural Organisation (FAO) Ethiopian Highland Reclamation Study (FAO, 1986) comes closest to reducing this uncertainty. Other more generally argued comments (e.g. Lamb and Seifulaziz, 1983; Timberlake, 1984; Independent Commission on International Humanitarian Issues [ICHI], 1986a, 1986b) link hunger to its 'real' cause—land degradation. These statements are useful because they help to mobilise international opinion, but they tend to argue from generalised statements and have to look to a few case studies scattered around the continent for empirical support. Subsequent research will have to be more rigorous, involving better measurement and a more refined methodology, to be useful in trying to solve specific problems.

Stocking (1978, 1987) has admirably summarised the problems of the measurement of land degradation and it is not the purpose of this article to rehearse them in any detail. There are many cases of estimates of rates of soil erosion for the same areas which vary by a factor of 50 and some which are clearly quite ludicrous (see, for example, Seckler, 1987: 86). Figure 1 illustrates the degree of uncertainty which surrounds estimates of suspended sediment yield within Africa (Stocking, 1987: 54). While it must be admitted that there are alternative routes to measuring land degradation other than using suspended sediment yields, measurement of this parameter is central to most attempts to measure rates of soil erosion.

One of the most important issues here is the relationship between soil loss and yield of reduction of crops and pastures. Stocking and Peake (1985) have

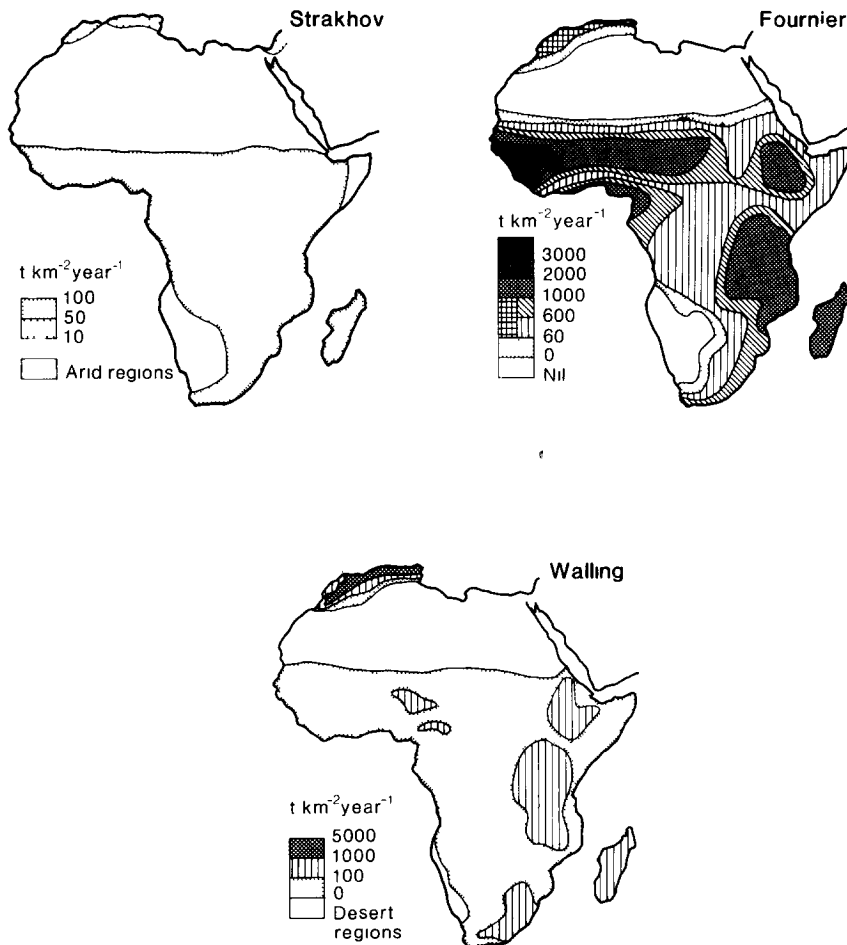


FIG. 1 *Maps of suspended sediment yields within Africa (from the work of Strakhov, 1967, Fournier, 1960, and Walling, 1984)*

reviewed the research that exists and have shown the enormous gaps in our knowledge. They point to research which clearly exposes the very great difference in impact upon crop yields of a given rate of soil removal between temperate and tropical soils, but these are measured in order of magnitude rather than on an accurate continuous scale of measurement. Quantitative modelling of the relationship between erosion and yield simply is not good enough to estimate future benefits of soil and water conservation on the base of general linear relationships. On-site data is also lacking to compare the situation with and without any proposed project. Differences in rainfall (amount and intensity), sunshine, hail, temperature, etc., clearly make one year's data on erosion and yields totally inadequate, and reliable time series information with these and other causal variables reliably measured simply does not exist outside the experimental station. Cross-sectional data, from which *ceteris paribus* assumptions can be safely allowed for to isolate the impact of a particular land use or conservation project, run into the same problem—these usually are not available (Bojo, 1986: 26, 27).

What is becoming clear from this research, however, is that productivity of many tropical soils is extremely sensitive to even small soil losses. Some Nigerian data suggests that yields were halved on 1 per cent slopes after only 30 tonnes/ha of soil loss, and it has been suggested that a tolerable rate of soil loss for such soils may be as low as 0.5 tonne/ha/year (Lal, 1984). Also, it seems that crop yields in many tropical soils fall in a negative exponential form with increasing cumulative soil loss. This means that initial soil losses have an enormous impact upon yields, and it is therefore tempting to put scarce resources for conservation *not* where soil has been eroded for some time and has low present productivity (and therefore yields low net benefits to conservation investment), but to put resources in areas of present high productivity not yet affected by soil loss but under threat from it. It follows that many already eroded areas may be beyond economic rationality where resources of funds, research and agricultural extension are as scarce as they are throughout Africa. However, all these trends need verification (Stocking and Peake, 1985: 28). Palliation of the data and modelling problems can partly be effected by the development of cheap and robust methods of measurement (e.g. Abel and Stocking, 1987).

An important element in the state of uncertainty about what is actually happening to the environment in the longer term (say, over one or two decades) concerns the difference in perceptions of land users and their governments. There are many examples of colonial officers writing in dismay in the 1930s and 1940s about the cultivation of steep slopes, land slips and impending catastrophe due to rapid population growth. Yet population densities two or three times those of half a century ago exist today in the same areas. Much of these colonial views can be laid at the feet of prejudice and a Eurocentric technical model of 'good farming', but also it seems that many African environments are more resilient than Europeans thought. Land slips, which suddenly appear in large numbers on a hillside, may prompt dire predictions from officials, but they are usually covered in vegetation again and are stabilised soon after (Berry and Townshend, 1973).

Also governments perceive the erosion problem in terms of their own technical assumptions. In much of colonial East Africa government soil

conservation officials carried with them technical training acquired in the USA or from textbooks written there. These sources strongly emphasised the role of the gradient and length of slope as the major determinants of soil erosion and therefore recommended terrace construction as the chief technique for soil conservation. It was not until after the Second World War that it was realised that for many parts of East and Central Africa, the characteristics of the slope were much less important in determining the rate of erosion than previously thought. Instead, the role of rainsplash is now seen to be very important, and therefore cover cropping and mulching are the appropriate conservation techniques (Anderson, 1984). Indeed, intercropping and relay cropping are integral parts of many African farming technologies, and the coercive policies to get Africans to construct bench terraces are often unnecessary (Mlia, 1987; Blaikie, 1987).

A second area of uncertainty is the much contested attribution of perceived changes in erosion rates to human agency or to climatic change. The 'desertification issue', so intensively written and talked about in conferences after the Sahel famines of the early seventies, revealed unbridgeable gaps in interpretation. In some localities there does seem to be strong evidence for desertification brought about by changing land-use practices (e.g. Lachenmann, 1985, for the Sahelian zone; Sinclair and Fryxell, 1985, and Lachenmann *et al.*, 1985, for Mali; and Franke and Chasin, 1980, for southern Niger). However, there still is considerable uncertainty over whether the desert is advancing and, if it is, whether human agency is or was responsible. Part of the debate is rehearsed by Watts (1986: 8), who writes: 'a good deal of the discussion of Sahelian degradation remains hopelessly entangled in unproven generalities and dubious, though unstated, assumptions concerning the ecology of grasslands, patterns of overgrazing and agronomic practice'. Also any attempt to attribute desertification to 'natural forces' other than the penetration of capitalism is written off in some quarters as merely a bourgeois red herring. This is a simplistic and unnecessary polarisation since it is the dialectic between environmental and social change which must provide the context in which land degradation is discussed.

The third source of uncertainty is more relevant to the concerns of this article: does land degradation matter and, if so, to whom? This question has already been discussed by Blaikie (1985: 12–37) who argues that the key issue is the prospect of future access of farmers and pastoralists to resources and technologies which can either halt or compensate for land degradation. Seckler (1987) and Nobe and Seckler (1979) have used the familiar economic notion of option values, which is helpful here. An option value is like an insurance policy: it is a premium which one is willing to pay over and above the normal cost of an operation in order to be able to reduce risk and uncertainty. In the field of soil and water conservation it may simply not be known how much, if anything, a certain amount of soil at a particular place will be worth to future generations. It may be known that without conservation the soil will be eroded and the option of future use will be lost. If it costs a farmer or a project something to preserve this option, it is useful to know what the value of this option is. In the case of Lesotho, for example (Nobe and Seckler, 1979), the costs of soil conservation are high and the return to conserving already damaged soil of poor present productivity rather low. At

present many Sotho work in South Africa as wage labourers. However, what happens if the Sotho are sent home and have to rely again on poorly rewarded agricultural and pastoral activities? The uncertainty in this case is acute, and therefore to reduce it and to ensure the physical basis of an alternative means of livelihood back home will be worth a great deal. Therefore it is even more worthwhile spending resources now on conservation, although the returns to agriculture are low and alternative employment for males still exists at present. None the less, whether option values can come to the rescue and provide a rationale for conservation sufficiently strongly to bring it about is open to doubt. As Nobe and Seckler see it:

... under present conditions, the productivity of agricultural land in Lesotho is so low that the economic conditions of a conventional conservation programme do not adequately exist. Agricultural practices are such that, even in the best possible conservation works, the land and the works themselves will continue to erode. Secondly, the agricultural production of land is so low that it is questionable if the net returns are sufficient to pay for even the maintenance costs of the conservation works—much less their total costs. [*ibid.*: 152]

Therefore the way in which the notion of option values is used may be more effective in justifying government action in soil and water conservation than as a normative or empirical model for individual behaviour. Indeed, Turner (1982) has pointed out in the Lesotho context that the appearance of a gully in a farmer's land substantiates the farmer's view that it would be more worthwhile to go to the South African mines rather than to expend ill-afforded resources in patching up land of very low productivity. There is a reinforcing element here which obscures the option value from the individual and yet at the same time enhances it. There may be a case for extending this notion to a more comprehensive calculation which takes into account the prospects of future farmers and pastoralists. These include access to labour, capital and technologies which would serve either to mask the effects of soil erosion (e.g. the application of chemical fertilisers, the use of improved varieties of crops which would make up for yield losses of non-improved varieties due to soil erosion) or to provide alternative incomes from sources outside agriculture. There are possibilities for assessing longer-term trends of national or regional agrarian economies using a combination of aggregate statistics and political economic analysis. Such an exercise would reveal in much of Africa that very large numbers of farmers and pastoralists will not have access to new farming technologies or the resources to develop or even to continue to use their own, and will be boxed into existing, eroded lands without a significant chance for alternative employment. Option values in the face of population growth, uncertain off-farm employment opportunities and differentiation would seem to be very high in these cases (for example, see USAID, 1981, for Ruanda; Roder, 1977, for Swaziland). However, this type of informed scenario building with some sort of probabilities attached moves the focus of what is strictly meant by an option value to one of probable future income streams from agriculture and pastoralism.

The last source of uncertainty concerns the existence of multiple levels of abstraction and generality in the analysis of land degradation. This is the 'levels and scales' premise of the JCAS project outlined above. Land

degradation occurs at the juncture of human activity and a given site with particular physical characteristics. It is caused by repeated decisions on the use of land which derive in turn from the agrarian political economy. Therefore there are general processes and relations between land users of different classes, other 'non-rural' classes, the state and the world economy which determine these land-use decisions. These occur at different levels of specificity and often have no geographical manifestation—they do not occur at specific places at all. However, land-use decisions and the nature of the land where they are taken are place-specific. Confusions arise when generalisations are made about the causes or impacts of land degradation (particularly resulting from single hypothesis explanations) which do not differentiate between general social processes or change (e.g. population growth or proletarianisation) and the place-specific occurrence of degradation, that is, where the resulting land degradation actually occurs and has its immediate impact.

Moving now to uncertainties in research which focus upon access to resources, the contributory factors are rather different and the levels of uncertainty are perhaps lower. First, the problems of measuring agrarian change, although considerable, are of a lesser order than those which face the measurement of land degradation and decline in yields. Even here there are still formidable problems, as discussed by Berry (1984a: 61–2), particularly with aggregate agricultural production data. None the less, there are a number of fairly reliable surveys which give good indications of changing control of access amongst different groups. The Zimbabwe National Household Capability Surveys (ZNHSCP), the Botswana Rural Income Distribution Survey of 1974–75, the 1980 Agricultural Census of Malawi, the Kenya Integrated Rural Surveys, or smaller *ad hoc* studies (e.g. Marter and Honeybone, 1976, in Zambia) are a few examples of data sets upon which a fair degree of confidence can be placed. They are, however, only empirical descriptions, with in-built categories and choice of variables, and describe a situation at one point in time. The question of process and change, of course, is much more open to debate, and here it is quite clear that there are a large number of theories and models of change, but they do not directly address the environmental and social problems addressed by this article. There is a well-developed literature on the differentiation of the African peasantry (see as a small sample of this genre *Review of African Political Economy*, 10 [1977] and, for Sudan, 26 [1983]; Bernstein, 1977, 1979, on the 'reproduction squeeze in Tanzania'; Franke and Chasin, 1980 for the west Sahel; Watts, 1984, for northern Nigeria; Marter and Honeybone 1976 for Zambia; Shepherd 1984, for Sudan). Hardly anywhere in this very large body of literature is there an accurate and systematic treatment of the relationship between changing patterns of access to resources and environmental change. There may sometimes be passing mention of the environment but it is usually unsubstantiated, although Richards (1983: 23) has also drawn attention to the rift between agrarian sociology and agricultural ecology. There is also another well-developed body of literature on farming systems in Africa, often with explanations offered for observed changes. Sources such as Allan (1967), Ruthenberg (1983), and the continuing programmes in cropping systems by IIRI and ICRISAT have thrown much light upon some of the important

changes in agricultural practice. However, these studies by and large have not sought to explain these changes in terms of social processes, particularly the mechanisms which distribute access to resources among a rural population. Below some suggestions are made about a possible cross-fertilisation between agrarian sociology and agricultural ecology.

APPROACHES TO REDUCE UNCERTAINTY

Thompson and Warburton (1985), when faced with 'uncertainties on a Himalayan scale' surrounding what was to be done in that region, suggested that we perhaps concentrate less on the question 'What are the facts?' and more on 'What would you like the facts to be?' In a series of articles in *Mountain Research and Development* they proposed some solutions. Some of these can be adapted to the circumstances of land degradation research which share this degree of uncertainty. It may be worthwhile to accept the 'great plurality of problem definitions, perceptions, expectations and rationalities' (Thompson and Warburton, 1985: 123) and to make a starting point with the definitions, perceptions and so on of the land users themselves. This view converges with at least three contemporary approaches to agrarian issues. The first is the recognition and re-evaluation of indigenous technical knowledge of African farmers and pastoralists (see, for example, Allan, 1967; Prothero, 1972; Richards, 1985). This has partly come about because of a realisation of the futility of one (usually imposed) perception of agricultural research to solve one narrow set of (usually settler and commercially orientated) problems (for example, Anderson and Millington, 1986, for anglophone Africa; Stocking, 1984, for 'colonial' Africa; Robinson, 1978, and Klepper, 1980, for Zambia; Eltringham, 1985, for Uganda; Millington, 1985, for Sierre Leone; Beinhart, 1984, for South Africa; Berry and Townshend, 1973, for Tanzania). Partly too, it is recognised that lack of access by peasant farmers to new agricultural technologies is the other side of the coin of 'inappropriate technologies'. This debate can be followed in Richards (1985), Ruttan (1981, 1982) and Biggs (1983, 1984). This new assessment of existing technologies must allow for variety in objectives and access to resources on the part of land users and, of course, variety in the resilience and sensitivity of the land itself. Intercropping is a case in point: Richards (1983: 26) has pointed out that this group of agricultural techniques has the advantages of effective soil and moisture conservation, spreading labour demands across the agricultural calendar and reducing the risk of pests, plant diseases or drought. However, they are inimicable to mechanisation and established research design on agricultural research stations and have systematically been denigrated and discouraged from colonial times to the present day (Belshaw, 1979).

The second contemporary approach with which this article converges is the increasing emphasis in improving productivity of farms and in soil and water conservation projects on 'starting where the farmer is' and integrating conservation with existing farming or pastoral practice. 'Policy can only succeed if it is implemented by farmers themselves, which means they must be trained to build appropriate adaptations into their existing environmental knowledge and skills' (International Food Policy Research Institute, 1985: 7). Other research findings emphasise that conservation can only be

successful by integrating it into on-going farming practices (e.g. USAID, 1985, in an evaluation of two conservation projects in Somalia and Ostberg, 1986, for a reasonably successful project in Tanzania). Thus attention has been increasingly focused on the particular circumstances of different land users 'in place'—in their specific natural and social environment. Single solutions and invariable explanations and policies are therefore increasingly recognised as erroneous and harmful.

A third converging approach to the understanding of agrarian societies in lesser developed countries centres on the 'new' household economics and, related to it, the issues of intra-household differentiation in work, control, decision making and access, particularly those deriving from gender and age-based seniority. Land degradation studies have seldom, if ever, considered the issue of conservation to be more than one considered by an unspecified 'head of household'. In most of Africa decisions on production and consumption are taken by a number of different people. Responsibilities overlap or sometimes contradict, between men and women, between elders and juniors, and so on. From earlier theoretical developments applications to Africa have been made (e.g. Low, 1986) and it is clear that farm decision making can sometimes much more satisfactorily be understood in terms of intra-household comparative advantages of household members in generating income, and of the politics of how this advantage is structured. There seem to be tremendous advantages for land degradation research to recognise that conservation usually cannot be conceptualised as the product of the decisions of an unidentified (but implicitly a single male) head of household (Guyer and Peters, 1984). For example Richards (1986) likens the 'farming venture' undertaken by the Mende of Sierra Leone to a ship sailed by a crew hired and paid off at each point during the voyage. Each voyage takes place over a catenary profile including different soils and moisture conditions, involving the labour of women in some areas, senior women in others, men for certain agricultural activities at certain times, and so on. It is only at a particular point in the agricultural calendar that anything approaching a discrete 'farm household' appears at all. Decision making is specific to particular places but is also linked on certain issues with the farming community. Guyer and Peters (1984) have edited a number of articles which identify variously 'hearth-holds' (matrifocal subdivisions of a patri- or matri-centred household [Ekejiuba, 1984]) and other subunits, usually gender specific, sometimes fluid and temporary, which control certain types of land and the production of specific crops. For example, Schlippe (1956; quoted in Ruthenberg, 1983) identifies specific responsibilities of the first son, second sister, mother, husband, etc., in a cultivation of different plots of land in a Zande holding, and there are so many examples throughout sub-Saharan Africa that it may be said that a division of labour not only between agricultural tasks but also between spatially defined units of land is the rule rather than the exception. This recognition also directs the explanation of land degradation 'downwards' to the socially and environmentally specific, even to below the 'household' (which in any case has been claimed to be part of a Eurocentric projection rather than an African reality).

In summary, three converging approaches to the research into agrarian issues in developing countries have been identified (plurality of perceptions

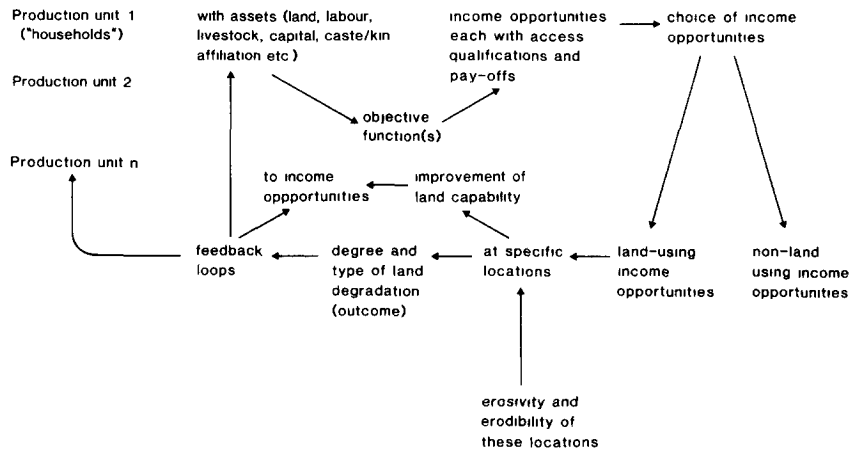
and problems; the need to root policies to improve farm production in the social and natural environment of the farmers themselves; and a focus on agents below or outside the 'household' level). All approaches suggest that resource use and environment relationships are highly conjunctural and that grand single theories usually fail. We therefore need an approach which can theorise this conjuncture; which allows single hypotheses to operate but in varying combinations; and yet which can combine the natural characteristics of land and vegetation at one point in time with cumulative land-use decisions which come about through more generalised and larger-scale processes in agrarian society as a whole.

A RESOURCE-ACCESS MODEL FOR LAND-USE DECISIONS

This section outlines an approach which is a research agenda with an implicit theoretical basis. It has already been outlined by Blaikie (1985) but is developed and modified here to provide an essential basis and input into material presented below. Its purpose is to explain the relationship between agrarian political economy, land management and use, and environmental change. It is not a model which analyses the agrarian political economy itself, although one of its central functions is to trace through the impact of that political economy on land use and environmental change. Attached to various components of the models are specific issues on which theoretical debates have centred, and some of those central in African political economy today are briefly discussed and linked to the model. In this way the model provides the 'spine' of an approach to changing resource access to which theoretical and empirical material can be attached.

The model starts by identifying what we can initially label 'production units' (which may be peasant households, 'hearth-holds', pastoral extended groups or commercial farms). The extent to which these units will have to be disaggregated will depend upon the degree to which land-use and land-management decisions are taken by separate individuals within the unit. Each product unit has an array of assets both physical and human (for example, the right to cultivate land of a particular type, labour and tools; education; affiliation to a dominant tribe or influential kinship network) and the distribution of these assets and resources will largely be determined by class and gender relations.

Each unit views income opportunities to which are attached access qualifications. Many of these are simply different cropping systems, each with a requirement per unit land of labour, seed, purchased inputs, etc., but others will be non-agricultural. Each opportunity has an expected pay-off (e.g. crop yields, money paid per hour, profits to a business concern). The array of income opportunities open to any group of farmers or pastoralists will reflect ecological conditions, the level of technical development in agriculture, and levels of economic and social development in the region. In Lesotho these opportunities are concentrated for males in wage labour in South Africa and for females in lowly rewarded agriculture or in government service. In the more remote and thinly populated areas the range of opportunities is quite different and consists of a wide range of crops grown in shifting cultivation systems, sometimes offering a relatively high productivity

FIG. 2 *Summary of resource-access model*

of labour, but with very few non-agricultural opportunities. The choices which farmers make between different cropping systems and between different crops at a particular point in time need only be modelled for these purposes to the extent that (a) specific income opportunities are contributing to land degradation and (b) access positions are sufficiently differentiated in an area to preclude groups from taking certain management decisions which may reduce or halt that degradation.

How the production unit approaches these opportunities and prioritises those open to it is, of course, a central subject of agricultural economic research, from Chayanov (1966), Lipton (1968), Weeks (1970), Binswanger (1980), Evanson (1981) and many others. Whatever objective functions seems most appropriate to the particular unit, the income opportunities are taken up, some land using (and of interest to the analysis of land improvement or degradation) and some not. These decisions are then located on a map at sites which have varying erodibility and forces of erodibility acting upon them. These 'natural' characteristics of land are rainfall (particularly its intensity and seasonality) and land-use decisions which together determine the degree if any of land degradation. Feedback loops from the 'state of the land' to the structure and assets of the production units, the income opportunities and their pay-offs complete the circuit. Some of these feedback loops can be very important indeed in the future history of social-environmental relations. Land capability may be improved by enhancing its nutrient status or investing in bunds or terraces, which will raise the future income of production units. Or it may decline, which will reduce incomes and bring about other changes in the objective functions of the units involved (for modelling of these responses, see below). Typically, environmental decline raises the costs of the reproduction of the productive unit or 'household'. It is usually women who will have to travel farther to collect firewood (Ki-Zerbo, 1981) or water for domestic purposes, and thus labour will be withdrawn from other agricultural operations.

In this model the agrarian political economy defines the 'rules of the game'. As we have already indicated, the social relations of production in a particular region define the distribution and access to resources as well as the flows of products or profits. In so far as the politics of the household, revolving around gender and age, may be decisive in land-use decisions, the 'productive unit' will have to be defined in a more disaggregate manner. Access to land will be governed by particular types of land tenure—common grazing lands, various legal-cum-administrative models of settlement (e.g. 'models' A to D in Zimbabwe), and various degrees of state intervention (often in conflict with local chiefs) in the allocation of land. Prices for agricultural outputs and inputs may be fixed by parastatal boards or by merchants acting in a free or restricted market, and so on. Relative prices and terms of trade between necessary purchased commodities for consumption and agricultural inputs on the one hand and primary agricultural outputs for sale on the other constitute another issue discussed by Bernstein (1977, 1979) in which the 'reproduction squeeze' raises the costs of production and forces the peasant to intensify commodity production, leading to the exhaustion of land and labour. So far this is a hypothesis, and it remains to be seen under what circumstances these terms of trade really do encourage poor land management leading to degradation. Most of the familiar issues of the African crisis (summarised in JCAS, 1986b) impinge upon those who make land-management and land-use decisions. In any particular circumstance a combination of these issues will be relevant and must be 'operationalised' in terms of the model (in the specification of distribution of resources, access qualifications for particular income opportunities, pay-offs, etc.)

It is necessary to chart the impact of changing agrarian political economy in terms of precise land-use decisions at particular locations in order to trace the impact of changes of the broader political economy upon land degradation. Satellite imagery, aerial photographs, land-use surveys and topographical maps are important resources in this type of research. Otherwise crucial aspects of differentiation of producers, commercialisation, population growth and other fundamental changes cannot be considered simultaneously and traced through the circumstances of people onto the land itself.

In the following section this dynamic model provides a point of entry for understanding the decision to alter farming and pastoral practice and to conserve soil and water by changing land-use practice in response to land degradation. As will be explained, access to resources is crucial in these decisions too.

RESPONSES BY LAND USERS TO LAND DEGRADATION

In the previous section a recursive model of agrarian change, land use and (possible) land degradation was discussed. Clearly we have to ask why sometimes degradation continues unchecked and in other cases successful modifications of farming practice take place. As the previous section explained, single hypotheses are often put forward but most of these prove to be conditional hypotheses only. Instead, attention could perhaps be focused on the detail of decision making in the face of a decline in the capability of the land and/or in the face of other challenges such as population growth. When

these hypotheses are considered in a broader context in which farmers or pastoralists operate, they may frequently be additive, cancelling or mutually modifying in each particular case. This section explains first the steps of the model in outline and then some of the substantive and theoretical issues which attach to each step.

Figure 3 outlines the model. In box 1 are listed the social and environmental data at time t which together form the context and desiderata for land management practice (box 2). The data consist of socio-economic characteristics of the decision-maker(s) (1, points 1–5).

We have already indicated the range of political economic processes which work through the relations of production, the operation of the market and so on, and these processes provide the dynamic and explanation for a static representation of this set of socio-environmental data at a particular moment in time. A distribution of land use and management across production units and over a differentiated geographical area (of different soils, slopes, vegetation, etc.) results from these desiderata. Also there are a number of issues in the forefront of agrarian and political economic debate in Africa which relate to these desiderata. The first is the knowledge and skill of farmers and pastoralists in relation to conservation techniques. Why, it may be asked, is there claimed to be such widespread degradation in Africa when farmers are reputed to have such a fine-tuned knowledge of the environment? Accounts of farmers' land-use strategies and knowledge in Allan (1967), Brokensha, Warren and Werner (1980), Watts (1984), Richards (1985) at first sight are difficult to square with reports of desertification and increasing erosion. One reason may be the prevalence of part-time off-farm and wage employment of land managers (or the 'peasantariat', Reynolds, 1986) which alters radically the amount and quality of indigenous knowledge which is essential for them to have. It may be simply that people only learn about what is important to them. Use values, even in materially simple societies, are multiple and to obtain them from nature requires detailed knowledge. As soon as the market replaces use with exchange values, parts of the household economy revolve around purchased inputs and it is no longer necessary to know of, for example, the healing properties of plants (pharmaceuticals are purchased instead) or of the illnesses of cattle (professional veterinary advice will do). If much of the household economy revolves around wage labour rather than agriculture, knowledge about household economics and not agriculture becomes more appropriate. For example, the knowledge of the Batswana in southeastern Botswana about their pastures and livestock was poor and their perception of undoubted pasture degradation rather limited. However, they were concerned and knowledgeable about maintaining their household income and shifting household labour around between activities (Abel, personal communication). The neighbouring Bushmen, on the other hand, are still extremely knowledgeable about their environment, since they still require multiple use values from it. Could it be that a market for labour deskills many farmers and pastoralists at the same time as offering alternative means of securing a livelihood.

To return to the next step of the model, the resilience and sensitivity of the biome (3) determine the degree to which the capability of the land is maintained or not in the face of these practices. Thus 3 refers to a purely

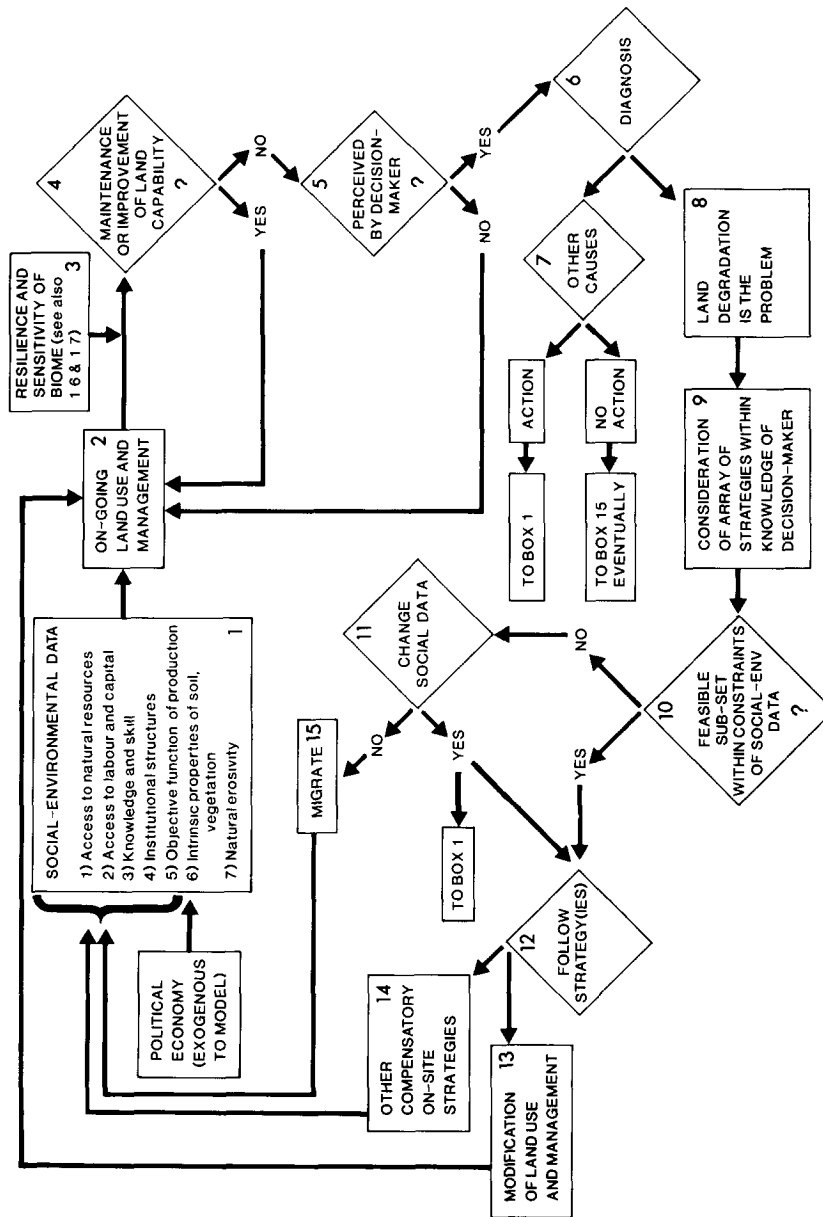


FIG. 3 Decision-making in land management

'place-based' consideration of how land stands up to on-going use and management. If the land does, and productivity is maintained (4), land use will have no reason to be altered on account of degradation alone. However, if land capability is declining, then the question arises whether decision makers perceive it or not (5). Hudson (1981) says 'my experience is that the small farmer has little idea of how much soil he is losing', while other commentators report an intimate knowledge of what is happening. Richards (1985) for West Africa, Berry and Townshend (1973) for Tanzania, and IFPRI (1985), which lists many varied conservation methods from the whole of sub-Saharan Africa, all offer clear examples of the latter state of affairs. If degradation is *not* diagnosed, on-going and unchanged practice will continue, until presumably, perception of the decline is inevitable. If yes, then a 'diagnosis' is made (6), and it is important to specify the basis on which it is made. Land degradation may be taking place, with the result that livelihoods are being threatened, but it may be the latter fact rather than the former which is diagnosed, and causes other than land degradation (7) may be identified. Homewood (1985) affords an excellent example in the Baringo District of Kenya. The plateau and hill area surrounding the Baringo swamp have long been deteriorating due to overstocking and physical erosion, and this is clear for all to see. Foreign advisers diagnosed the problem as pasture degradation caused by overstocking, and have proposed destocking, group ranches and so on, but without much success. The Il Chamus pastoralists do not perceive the cause of their parlous state to be erosion, and a reduction in the capability of the dry land areas on the hills and plateau was, in their view, beside the point. Most of the Il Chamus cattle are based in the swamp area for most of the year and only move away onto the eroded uplands for short periods. The swamp itself supports grasses which are virtually impossible to graze out (e.g. *Cynodon* and *Echinochlon*, which are highly resilient). Therefore the main land resource on which they depend has not declined in its capability. There is undoubted overstocking, but the Il Chamus identify the problem as lying in the differential ability of the various social groups using the swamp to rebuild their stocks after the drought-induced losses. Absentee cattle owners, often with office jobs and other sources of income, are able to purchase cattle quickly, whereas local people cannot, with the result that the Il Chamus actually own today far less cattle than twenty years ago—hence their poverty. Another example of a diagnosis which did not identify land degradation as the problem was made by the Kikuyu in the 1950s when faced by land degradation and coercive conservation policies imposed by the colonial authorities. The problem was diagnosed as one of the alienation of land by settlers, leading to instant 'overpopulation', and the political solution of ridding themselves of the British appealed more than costly conservation works.

Mention was made above about different perceptions of environmental degradation. In this model it is the perceptions of the land users themselves which are being investigated. The divergence between those of the land users and those of the government is partly explained by some of the paths taken by the former that lead to outcomes of continuing land degradation. However, in many instances it is simply not clear whether, in cases where there is no major divergence in definitions of land degradation, it is taking place or not.

In other words, if farmers in the Communal Areas of Zimbabwe were to revisit their land thirty years from now, would they diagnose land degradation, using their own perceptions and definitions? The evidence is often contradictory. Sandford (1982) has reviewed the evidence from that country and concludes that it is still uncertain. While it is obvious that important vegetational changes have taken place there (West, 1968; Wilson, 1986) and erosion seems to be occurring (Campbell *et al.*, 1986; Kelly, 1971, 1973; Whitlow, 1986) it is also the case that both primary production (grass) and secondary production (livestock) have never been higher on the grazing lands of the Communal Areas (Abel and Blaikie, 1988). Is this rate of offtake sustainable or not? Probably not, in the case of Zimbabwe (Kelly, 1973), but that tentative view must be re-examined where the rainfall is even less, as in Botswana. There, according to Biot (1988) there will not be an erosion problem in the foreseeable future (some hundreds of years). So, perceptual differences of erosion apart, it is usually difficult to assert unequivocally that farmers do not appreciate the amount of soil they are losing (to adopt Hudson's, 1981, expression). We simply cannot substantiate the 'objective' view—that which is defined in terms that could be shared by land users and government/'outsiders' alike.

To return now to the perceptions and diagnosis of the land users themselves and their path of decision making, if the diagnosis that land degradation is the cause of a decline in land capability (8), an array of strategies which lie within the knowledge of the decision maker are considered (9), and a feasible subset is identified (10). This process will depend upon a number of factors in the data set (1). Here the access profile of the decision-making unit is crucial. Many conservation techniques are land-absorbing (see Millington and Mutiso, 1986) and in densely populated areas with restricted access to new land this factor may put bench terracing and grass strips beyond the reach of many people. Others are labour-intensive and the opportunity costs of labour may well be too high in many African cases. Changes in agronomic practice in terms of altered site selection, soil transfer, timing of planting, crop choice or the introduction of mulching may be the only feasible solution. In Tanzania Berry and Townshend (1973) identified many such changes which were all made subject to the patterns of resource access of farmers—while others introduced by government demanded resources which the farmer considered scarce and these practices were not adopted.

At this point it is important here to specify carefully the field of choice of the decision maker(s). It is seldom a field-based cost-benefit consideration, nor even one based on the sum of agricultural/pastoral operations (or the 'farm'). Therefore it is one that differs fundamentally from full-time farmers who have exclusive control over their fields on which much Western agricultural economic treatment of conservation decisions rest. In much of Africa these decisions tend to be made in the context of overall income and leisure of a wider collectivity than merely a farming one. Particularly in Southern Africa, changes in farming or pastoral practices and related conservation decisions must take into account labour requirements which may find higher and more certain rewards in wage labour elsewhere, and which may not be available for the more labour-intensive soil conservation

works (IFPRI, 1985). It is therefore a comparative cost-benefit analysis that land managers use in assessing conservation that relates to the household economy rather than to land use alone. In many other parts of Africa too labour migration represents an income opportunity which strongly but indirectly affects land use and land management decisions. Clearly there is enormous variation (Freund, 1984: 21), from cocoa tree planting migrants in West Africa (Hill, 1963) to the plight of the black homelands of South Africa (Lipton, 1980), from the massive male outmigration of Lesotho (Murray, 1981) to impoverished pastoralists with alternative employment opportunities in Botswana (Abel *et al.*, 1985). Elsewhere in this series of articles Berry has suggested that instability and recurrent food dearths have contributed to a situation where:

People are reluctant to tie up capital in fixed assets or long-term projects and prefer to hold commodities . . . or specialise in activities with rapid turnover rather than in those with long gestation periods. . . . So that the quantity of commodities and services in the economy remains constant or declines, while their 'velocity of circulation' increases.

The implications for the commitment of present resources to effective land management for long-term enhancement of agricultural or pastoral income can easily be imagined. Further, many marginal areas which have become labour reserves for industrial or agricultural enterprises elsewhere have suffered from environmental decline, which in turn makes the comparative returns to investment in conservation even poorer (for a cross-country comparison see Blaikie and Brookfield, 1987: 113–20). Also there may not be the institutional structures to ensure co-operation between farmers or pastoralists, without which a particular strategy carried out by an individual land manager will not be successful. The question of the maintenance of effective common property resource management systems is relevant here (e.g. Abel *et al.*, [1985] for Botswana; Sandford [1983]; Behnke [1985]) (2). If no feasible subset of strategies can be identified, it may be possible to alter the social data (11) to enable a strategy to be adopted, and the extent to which this is possible will depend upon the economic and political power (and ultimately relations of production) of the land manager. For example, landlords and, in the past, feudal aristocracies and colonial authorities compelled their serfs or workers to construct terraces or plant trees (e.g. in Ruanda-Burundi under Belgian colonial rule). Alternatively farmers may seek to enhance their knowledge and skill from extension agents (if available), or form co-operative teams for conservation works, pasture regulation, forest planting and so on, where they did not exist before. If none of this is possible, land capability continues to decline and at least some land managers will be forced to leave (15).

If feasible strategies can be identified, then they are followed (12) and modifications of land-use practice and management take place (13), which are, in turn, inputs into ongoing land use and management (2) at time $t + 1$.

It is important to emphasise that the social and environmental data (summarised in box 1) are themselves in a state of constant change; conceptually they are the output of the resource-access model outlined above. For example, changes in the population of humans and/or livestock will

probably require intensification of farming, reductions in fallow, increases in the flow of imported nutrients (e.g. agricultural manure, forest litter, chemical fertilisers). The ongoing process of the conversion from shifting cultivation, fallow systems through to permanent cultivation as is occurring in many parts of Africa is a case in point and provides a continuous challenge to land management and, as it were, an exogenous impulse for change in box 2. Any outcome in control over resources or in soil and water conservation are crucially determined by the political economy of the agrarian society involved. The process of intensification of land use (as a result of such forces as population growth, market opportunities and the marginalisation of weaker groups), can now be viewed as one determined by the pattern of access to resources. While intensification may result from altered access patterns (and in some cases reduced ones), it also demands a new set of resources. Sometimes the people involved simply do not have them. Access to nutrients is a central factor here and one which prevents many from achieving a successful agricultural transformation, and which locks them into a cycle of untreated land degradation. The interface between pastoral and agricultural peoples is also important and there are many cases of the disruption of symbiotic relations between pastoral groups, which provide manure, and agricultural groups, which provide grazing (e.g. Scott, 1984).

In conclusion, this article has argued for a more 'place-specific' approach to the issue of land degradation which can at the same time pick up both the major changes in agrarian political economy which operate at national and international scales and the detail of 'household'/group decision making in land use and management. It is a research agenda which is based upon an overall model of changing control over resources and land-use decisions linked to another which identifies the decision-making process in the face of land degradation. It is a challenging agenda since it links political economy with quantitative modelling more associated with positivistic social science and normative agricultural economics. In my view this contradiction is a false one, since the models presented here and the theory behind data collection (categories, labels, choice of quantitative intervals, etc.) are sufficiently flexible to accommodate an ideologically explicit but quantitatively verified approach to social and environmental change in Africa.

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Résumé

Environnement et accès aux ressources en Afrique: un document programmatique

Le but de ce document est de fournir un programme de recherche sur l'environnement et l'accès aux ressources en Afrique. Il soutient une approche plus 'localisée' à la question de la dégradation des terres qui peut à la fois souligner les changements majeurs de l'économie politique agraire en vigueur aux échelles nationale et internationale ainsi que les détails de prises de décision collectives 'domestiques' en matière de gestion de l'utilisation des terres. Ce programme de recherche est fondé sur un modèle général de changement du contrôle sur les ressources et les décisions relatives à l'utilisation des terres qui est lui-même lié à un autre modèle qui identifie le processus de prise de décision par rapport à la dégradation des terres.

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